[INCH-POUND]
A-A-59221
May 18, 1998
SUPERSEDING
MIL-L-18947B
24 August 1964

COMMERCIAL ITEM DESCRIPTION

LAPPERS, POPPET VALVE, HAND POWERED

The General Services Administration has authorized the use of this commercial item description by all Federal agencies.

1. SCOPE. This commercial item description (CID) covers grade A and grade B hand-powered lappers of the oscillating, progressively or irregularly advancing type used primarily for lapping automotive engine valves and seats.

2. CLASSIFICATION. Lappers shall be of the following grades (see 7.2):

Grade A – Heavy duty.
Grade B – Light duty.

3. SALIENT CHARACTERISTICS.

3.1 Description. Grades A and B lappers are intended for use in maintenance which requires the lapping of valves and valve seats of automobile engines. They are satisfactory for valves and valve seats requiring a normal amount of lapping. Grades A and B lappers are not recommended for valve and valve seats which are badly pitted, burned, or scored unless the valves and valve seats cannot be replaced with new parts or that power tools are not available.

3.1.1 Grade A lappers. Grade A lappers are considered to be appropriate for use in maintenance shops which may encounter heavy work loads of valve and valve seat lapping during engine repairs. They are considered to have satisfactory service life of longer duration than grade B.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 15E2), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5120

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3.1.2 Grade B lappers. Grade B lappers are considered to be appropriate for intermittent use and are suitable for inclusion in toolboxes for maintenance of automobile engines.

3.2 Standard commercial product. The lappers shall, as a minimum, be in accordance with the requirements of this commercial item description and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this commercial item description but which are a part of the manufacturer's standard commercial product, shall be included in the lappers being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Material. The housing of the lapper shall be cast iron or drawn steel.

3.4 Design and construction. The lapper shall consist essentially of a housing, crank arm, valve lifting spring, extension drive stem, drive points, gear train or mechanism, ball thrust bearing and rubber suction cups. The lapper shall produce an oscillation of the valve around its axis with progressive or irregular advancement. Lappers shall be so constructed that no assembled part will became detached during normal use. All moving parts shall operate without binding or looseness that would impede proper function of tool.

3.4.1 Grade A. Grade A lappers shall be heavy-duty capable of withstanding 500,000 revolutions without permanent deformation when operated at 200 revolutions per minute with a torque load between 2 to 3 inch pounds (0.23 to 0.34 newton metre (N·m)) to the drive stem.

3.4.2 Grade B. Grade B lappers shall be light-duty capable of withstanding 90,000 revolutions without permanent deformation when operated at 200 revolutions per minute with a torque load between 2 to 3 inch pounds (0.23 to 0.34 N·m) to the drive stem.

3.4.3 Housing. The housing shall be of two half sections or one section with a cover. In either case, the half sections or cover plate shall be assembled securely to minimize both the loss of light density grease and the entrance of foreign particles. The assembly shall allow ready access for lubrication and removal and replacing of parts. The housing shall properly support the enclosed mechanism. A handle or knob, which affords a comfortable grip, shall be attached or integral with the top side of the housing.

3.4.4 Crank arm. The crank arm shall be of a length and shape to ensure clearance for hands in operating, and have sufficient strength to withstand intended usage. One end of the crank arm shall provide means for attaching to crank shaft. The other end of the arm shall have a handle (knob) affixed to rotate on a bolt, rivet or pin of the head and shoulder type.

3.4.5 Valve lifting spring. The lifting spring shall be helical, or hard drawn spring wire, oil-tempered wire or piano wire. The spring shall be 2.25 ± 0.12 inches (57 ± 3 millimetre (mm)) long by 1.0 ± 0.06 inches (25 ± 2 mm) in diameter. The helical string may be straight-wound or the ends may be conical in shape.

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3.4.6 Extension drive stems. Each lapper shall be furnished with a long drive stem not less than 7 inches (178 mm) long and not greater than 8 inches (203 mm) long. Each lapper shall also have a short drive stem not less than 3 inches (76 mm) long, and not greater than 4 inches long (102 mm). One end of the drive stem shall be threaded for attachment to the driving spindle. The opposite end of the drive stem shall be slotted and drilled for attachment of drive points.

3.4.7 Drive points. Drive points, in the amount and sizes normally furnished by the manufacturer, shall be included with the lapper. The points may be either one-piece universal blade type or individual units and shall contain a hole to allow for attachment to the stem with a cotter pin. When specified (see 7.2), special drive points shall be supplied.

3.4.8 Gear train or mechanism. Gears and pinions, if applicable, shall be secured to crank and drive shafts by a positive means to ensure proper alignment, rotation and meshing. In the grade A lapper, the relation between the spacing and number of teeth of the driving and driven gears or the capability of the operating mechanism shall be such that the sum total of the forward and reverse rotation of the drive stem will total between 450 and 675 degrees for each revolution of the crank. The grade B lapper shall conform to the above requirements exempt that the sum of both forward and reverse rotation of the drive stem shall total between 860 and 720 degrees for each revolution of the crank.

3.4.9 Ball thrust bearing. A ball thrust bearing shall be used to minimize friction and wear while the vertical drive shaft is rotated under pressure. Construction of bearing shall be such that the balls will not be disturbed when removing rubber suction cup from the drive stem.

3.4.10 Rubber suction cups. Each lapper shall be equipped with ten, vacuum type, rubber suction cups. Five suction cups shall be approximately 1.25 inches (32 mm) in diameter, and five suction cups shall be 1.75 inches (44 mm) in diameter. They shall be of conventional shape designed to force fit the extension drive stem or be secured by cotter pins to securely hold valves not equipped with spanner holes or slot in the valve head.

3.4.11 Lubrication. The bearings and gears shall be lubricated with light density grease to render the lapper ready for immediate use.

3.4.12 Weight. The weight of a lapper with a cast iron housing shall be 3.5 to 5 pounds (1.6 to 2.5 kg) for grade A, and 2 to 3 pounds (0.9 to 1.4 kg) for grade B. The weight of a lapper with a steel housing shall be 1.5 to 2.5 pounds (0.7 to 1.1 kg) for grades A and B.

3.5 Painted or coated housing surfaces. All surfaces of the housing normally painted as in manufacturer's commercial practice shall be furnished with not less than one coat of enamel or lacquer. Otherwise, surfaces shall be protected by a dense phosphate or oxide coating followed by a coating of oil, grease or compound preservative.

3.6 Corrosion resisting surfaces. All parts and surfaces, except the housing which are subject to corrosion shall be protected by cadmium plate, zinc plate, or by a dense phosphate or oxide coating, followed by a coating of oil, grease or compound preservative.

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3.7 Marking for identification. Wrenches and wrench assembly shall be marked with the manufacturer's name or readily identifiable trademark.

3.8 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with good commercial practice. Burrs, sharp edges, forging flash, and other harmful or extraneous material shall be removed.

4. REGULATORY REQUIREMENTS.

4.1 Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this commercial item description are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this commercial item description.

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of ASTM SI-10 (IEEE/ASTM SI-10), and all other requirements of this commercial item description including form, fit, and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING. The preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 The Federal Acquisition Regulation (FAR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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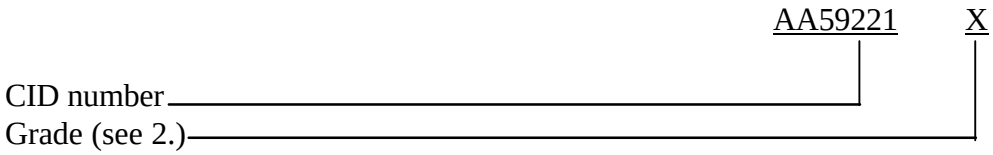
7.1.2 ASTM Standard is available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.1.3 IEEE Standard is available from the Institute of Electrical and Electronics Engineers (IEEE), IEEE Service Center, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this commercial item description.
b. Grade of lapper required (see 2.).
c. When special drive points are required (see 3.4.7).

7.3 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PIN to be used for items acquired to this description are created as follows:



7.4 Classification cross reference. Classifications used in this CID (see 2.) are identical to those found in the superseded Military Specification MIL-L-18947B.

7.5 Supersession data. This CID replaces Military Specification MIL-L-18947B, dated 26 February 1993.

7.6 Metric products. The values states in either inch-pounds or SI units are to be regarded separately as standard. Within the text, the SI units are shown in brackets. The values stated in each system are not exact equivalents; therefore, each system shall be used independently of the other. Combining values from the two systems may result in non-conformance with standard.

7.7 Subject term (key word) listing.

Hand powered
Seat valve
Valve lapper

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MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1

Review Activities:
Army - MD, SM
Navy – SH

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:
Navy - YD1

(Project 5120-0088)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-59221	**2. DOCUMENT DATE (YYMMDD)** 980518

LAPPERS, POPPET VALVE, HAND POWERED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME KENNETH A. POLLOCK	b. TELEPHONE *Include Area Code)* (1) Commercial (805) 982-5612 (2) AUTOVON 551-5612
COMMANDING OFFICER, NCBC CODE 15E2K 1000 23RD AVENUE PORT HUENEME, CA 93043-4301	

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